

From In-Sample Monotonicity to Out-of-Sample Returns

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ABSTRACT

The asset-pricing literature has proposed hundreds of factors that predict stock returns in-sample, yet out-of-sample predictability tends to decline (e.g., McLean and Pontiff, 2016). This paper shows that out-of-sample returns are predictable using in-sample factor “shape” characteristics—specifically, the monotonicity of factor portfolio returns across characteristic-sorted portfolios. Factors with more monotone in-sample return profiles deliver approximately 62.2 bps/month OOS return, while less monotone factors deliver about 22.3 bps/month. The evidence suggests that monotonicity helps identify persistent factors by disciplining spurious in-sample patterns arising from data mining.

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Imagine a journal editor receives a manuscript claiming a new return-predictive factor—validated with reported t -statistic using all available historical data. She must decide whether the finding is likely to survive out-of-sample. This is obviously a difficult decision—and arguably a billion-dollar question—because portfolio managers face the same problem when deciding which signals to allocate capital to.

Researchers have long recognized that many asset-pricing factors weaken once they venture beyond their discovery sample. McLean and Pontiff (2016) document that mean factor returns drop 26 % in out-of-sample tests and 58 % after journal publication. Using a multiple-testing framework, Harvey, Liu and Zhu (2016) show that 80 – 158 of 296 reported factors lose significance under standard corrections. Together these findings have shifted the discipline’s focus to a single overriding issue: can any purported risk or mispricing premium survive the new time periods? The robustness of factor predictability has thus become the central question in modern asset-pricing research.

Current responses to this problem remain useful but imperfect. One response, associated with Cochrane’s discussion of “factor fishing,” is to treat economically motivated factors more favorably. But implementing that principle in practice is inevitably subjective and difficult to standardize across studies. Chen, Lopez-Lira, and Zimmermann (2025) find that post-sample performance depends little on whether a predictor is peer-reviewed or data mined. A second response is statistical. Harvey, Liu, and Zhu (2016) argue that the conventional hurdle of $|t| > 1.96$ is too weak in a factor-zoo setting and propose a much higher cutoff—roughly $|t| > 3.0$ —for newly discovered factors.

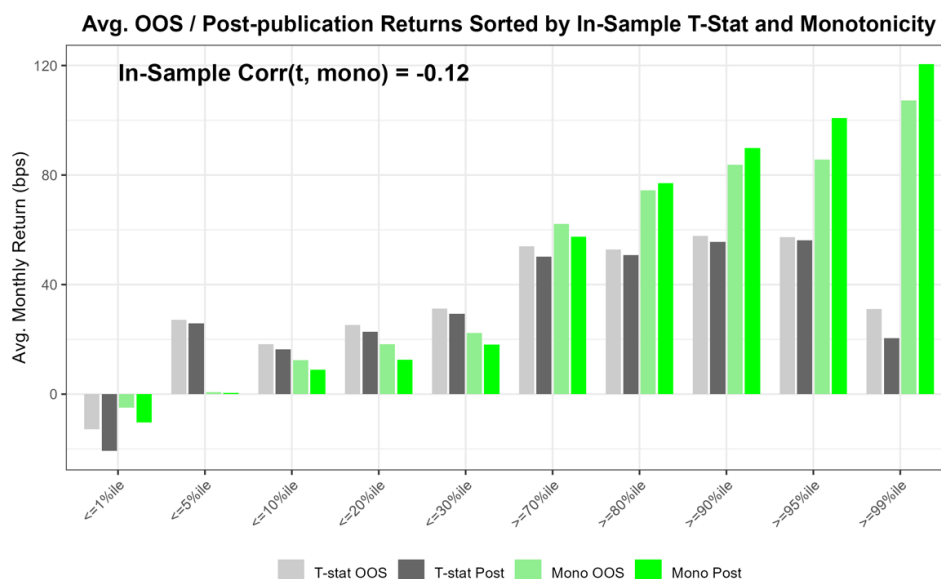
This paper introduces a new in-sample diagnostic: return monotonicity across characteristic-sorted portfolios. I measure it as the share of months in which decile returns rise (or stay flat) steadily from the lowest to the highest portfolio—an intuitive “fraction-monotonic” score. Applied to the 153 factors examined by Jensen, Kelly, and Pedersen (2023), this monotonicity metric strongly predicts both out-of-sample and post-publication returns.

Figure I summarizes the main results of this paper. At the cross-sectional factor level, both in-sample t -statistics and monotonicity are predictive of out-of-sample and post-publication returns, as average returns rise with stronger in-sample signals. Starting at the $\geq 70^{\text{th}}$ -percentile bin, however, factors selected by monotonicity already earn about 5 bps per month more than their t -statistics counterparts; the edge widens to roughly 10, 15, 20, and finally 70-80 bps in the $\geq 80^{\text{th}}$, $\geq 90^{\text{th}}$, $\geq 95^{\text{th}}$, and $\geq 99^{\text{th}}$ percentiles, respectively, producing a smooth step-wise ascent in the monotonicity bars versus the flatter, more erratic profile of the t -statistic bins. These widening gaps underscore that monotonicity is the stronger filter for identifying durable factors, and the low in-sample correlation between the two metrics ($\rho \approx -0.12$) further suggests they capture distinct dimensions of factor quality.

Figure I

Avg. OOS / Post-Publication Returns Sorted by In-Sample T-Statistics and Monotonicity

This figure plots the average out-of-sample and post-publication returns of factors. Each bin contains factors that are above or below certain percentile sorted by in-sample t-statistics or monotonicity. T-statistic is based on the top-minus-bottom factor return spread. Monotonicity is defined as the fraction of monotonic months over the total number of months. The in-sample, out-of-sample, and post-publication periods are defined by each factor's original sample period and publication year, respectively. The sample covers January 1926 through December 2024 and is based on monthly U.S. stock return data from CRSP. Accounting data are obtained from Compustat.



Although the t-statistic is intuitive and informative from both frequentist and Bayesian perspectives, it has one notable limitation. The standard top-minus-bottom t-statistic relies only on the two extreme portfolios, discarding information contained in the intermediate ranks and thus failing to fully exploit the cross-sectional structure of returns (Patton and Timmermann (2010)).

The monotonicity diagnostic offers several compelling advantages. It is non-parametric, rests on minimal assumptions, and produces a continuous score bounded between 0 and 1, making it easy to rank and implement. Unlike the conventional top-minus-bottom t-statistic, it exploits the entire decile return profile rather than just the two extreme portfolios. Moreover, a monotonic ordering of returns is not merely a statistical curiosity; it is a feature implied by a wide class of asset-pricing models. Together, these attributes allow monotonicity to pinpoint factors that reflect genuine mispricing or risk compensation while screening out spurious patterns born of data mining.

In this paper, I show, first, that in the cross section this measure strongly predicts both out-of-sample returns and post-publication returns, and is largely independent of the conventional top-minus-bottom t-statistic. Second, in the time series, the same result carries over to monthly portfolio performance when factors are selected based on their historical monotonicity measures. Meanwhile the in-sample t-statistic is itself a strong predictor of future returns, too. Combining monotonicity with the t-statistic in the monthly portfolio yields even stronger performance: factors that rank highly on both dimensions deliver the best subsequent returns.

Factors with high in-sample monotonicity earn stronger subsequent returns than factors with low monotonicity, although they also contribute more to portfolio volatility. At the same time, the profitability of monotonic factors is driven primarily by the long leg, while the short leg earns lower returns and is more vulnerable during periods of market stress.

Trading costs are a first-order concern in evaluating the post-publication profitability of factor portfolios. McLean and Pontiff (2016) show that post-publication returns are higher for factors concentrated in stocks with high idiosyncratic volatility and low liquidity, consistent with a trading-frictions interpretation. Against this backdrop, I show that the monotonicity strategy does not simply tilt toward harder-to-trade stocks. On the contrary, it tends to select stocks with larger mean and median market equity and lower average turnover over the sample period. This evidence suggests that the higher returns generated by the monotonicity strategy are unlikely to be compensation for higher trading costs.

My framework is subject to several limitations. In the baseline specification, a month is counted as monotonic only if decile returns are strictly non-decreasing across portfolios. This definition is intentionally simple, but it may be overly restrictive: months that are economically monotonic, in the sense that returns display a broadly increasing pattern with only minor local deviations, are nevertheless classified as non-monotonic. The resulting measure may therefore understate the true degree of monotonicity in factor returns. In addition, because strict monotonicity becomes harder to satisfy as the number of portfolios increases, the measure is sensitive to the choice of portfolio granularity. A useful direction for future research would be to develop a more robust measure that captures economically meaningful monotonic structure rather than strict statistical ordering.

Literature Review and Implementation

As the asset pricing literature has expanded into a factor zoo, concerns about out-of-sample decline and replicability have moved to the forefront. McLean and Pontiff (2016) show that factor returns fall by 26% out of sample and by 58% after publication. Harvey, Liu, and Zhu (2016) and Hou, Xue, and Zhang (2018) further question the credibility of many reported factors, arguing that significance based on the conventional hurdle of $|t| > 1.96$ can be misleading in the presence of extensive multiple testing. In contrast, Chen and Zimmermann (2021) show that most factors can be replicated in open-source data. Jensen, Kelly, and Pedersen (2023) extend this debate by recasting multiple testing in a Bayesian framework. They likewise find broad

support for factor replicability and argue that the out-of-sample decline in factor returns is an expected outcome of prior beliefs centered on zero alpha.

Asset-pricing research explains the out-of-sample decline of factor returns through three main channels: (i) time-varying risk premia caused by regime shifts (Lettau, Ludvigson, and Wachter (2006); Smith and Timmermann (2022); Guidolin and Timmermann (2008); Chordia, Subrahmanyam, and Tong (2014); Brogaard, Nguyen, Putnins, and Zhang (2023)), (ii) publication-informed arbitrage that quickly eliminates mispricing once a signal becomes public (McLean and Pontiff (2016); Brogaard, Engelberg, and Van Wesp (2023); Jacobs and Müller (2020); Bowles, Reed, Ringgenberg, and Thornock (2024)), and (iii) data-mining bias that inflates in-sample significance but fails replication (Linnainmaa and Roberts (2018); Chen and Zimmermann (2022); Chen, López-Lira, and Zimmermann (2025)). While the first two channels—time-varying risk premia and publication-informed arbitrage—reset expected returns and, in so doing, reveal how markets transition to a new equilibrium after fundamentals or information flow change, the third channel, data-mining bias, is within our control.

Much of the asset-pricing literature that seeks to curb data-mining bias still leans on statistical significance and marginal pricing ability. Harvey, Liu, and Zhu (2016) and Jensen, Kelly, and Pedersen (2023) develop multiple-testing frameworks to discipline false discoveries in the factor zoo. Green, Hand, and Zhang (2017) jointly evaluate 94 characteristics and show that only two remain significant after 2003; importantly, those are also the only characteristics that continue to generate meaningful returns thereafter. Giglio, Feng, and Xiu (2020) combine double-selection LASSO with Fama–MacBeth estimation to assess whether a newly proposed factor adds incremental pricing power after controlling for a large set of existing factors. This paper contributes to the literature from a different angle by focusing on a factor’s in-sample shape, namely the monotonicity of returns across characteristic-sorted portfolios, and showing that this feature strongly predicts future returns.

Literature has also considered monotonicity as a shape-based notion of robustness. Patton and Timmermann (2010) develop nonparametric tests for monotonicity in expected asset returns, and Romano and Wolf (2013) provide related procedures for testing monotone relations between characteristics and expected returns. While these studies establish the importance of monotonic portfolio structure, they are primarily designed for statistical testing, yielding reject-or-not-reject evidence rather than a continuous ranking signal. In addition, they have not been brought to bear on large-scale out-of-sample tests in the factor zoo. This paper takes a different approach by constructing a continuous measure of monotonicity and applying it in a broad factor-testing environment to study whether in-sample portfolio shape predicts future returns.

The paper proceeds as follows. Section I describe the data and methodology. Section II presents empirical results. Section III discusses how in-sample monotonicity translates into future returns. Section IV concludes.

I. Data and Methodology

Return data come from the CRSP U.S. monthly stock file, beginning in 1926, and accounting data come from Compustat. Excess returns are measured relative to the U.S. Treasury bill rate. To reduce the influence of data errors, I winsorize monthly stock returns at the 0.1th and 99.9th percentiles. I incorporate delisting returns from CRSP whenever available; when a delisting return is missing and the delisting is performance-related, I assign a return of -30%, following Shumway (1997). I construct 153 stock characteristics following Jensen, Kelly, and Pedersen (2023), including 94 accounting-based characteristics and 59 price-based characteristics. For the accounting-based variables, I assume that accounting information becomes available four months after the fiscal period end. When constructing valuation ratios, I use the most recent available price.

I build one-month holding-period factor returns as follows. First, in each month, I sort stocks on each characteristic into five deciles. For each decile, I compute the mean one-month-ahead return using an equal-weighting scheme. The factor is then defined as the top-decile return minus the bottom-decile return, corresponding to the excess return on a zero-net-investment long–short strategy. The factor is long (short) the decile identified by the original paper as having the highest (lowest) expected return. I also measure, for the traded stocks in the top and bottom deciles, the average market equity and total turnover—defined as the sum of the absolute changes in stock weights—for each factor in each month.

I require a factor return to be non-missing only if both the long and short legs contain at least five stocks. In addition, each factor must have at least 60 non-missing monthly observations to be included in the sample. The in-sample, out-of-sample, and post-publication periods are defined using the factor’s original study sample period and publication year, respectively.

A month is defined as monotonic if portfolio returns are non-decreasing across deciles. For each factor i , I define the monotonicity measure over a given period as:

$$Monotonicity_i = \frac{\# \text{ of monotonic months in the period}}{\# \text{ of total months in the period}}$$

Table I presents summary statistics for the key variables used in this paper. Panel A reports descriptive statistics at the cross-sectional factor level, while Panel B reports descriptive statistics at the factor-month level over the full sample period.

Table I

Summary Statistics

This table presents summary statistics for all 153 factors at the overall factor level (Panel A) and at the factor-month level (Panel B). Return, Long Leg, and Short Leg are reported in basis points. T-statistic is based on the top-minus-bottom factor return spread. Monotonicity is defined as the fraction of monotonic months over the total number of months. Size is measured in thousands of U.S. dollars, and Turnover is defined as the sum of

absolute changes in stock weights for traded stocks within a factor-month. The in-sample, out-of-sample, and post-publication periods are defined by each factor's original sample period and publication year, respectively. The sample covers January 1926 through December 2024 and is based on monthly U.S. stock return data from CRSP.

Panel A: Mean of 153 Factors Mean Across Sample Periods				
	Full Sample	In-Sample	Out-of-Sample	Post-Publication
Return	55.01	60.93	48.92	45.71
T-Statistic	4.42	4.18	2.01	1.63
Monotonicity	0.093	0.098	0.085	0.083

Panel B: Descriptive Summary at Factor-Month Level											
	Mean	Std. Dev.	Min	5th%ile	10%ile	25%ile	Median	75%ile	90%ile	95%ile	Max
Return	47.67	442.08	-2.2E+04	-514.23	-312.53	-107.42	46.11	208.15	433.97	635.47	1.0E+04
Long Leg	27.85	704.48	-9.5E+03	-998.54	-719.75	-334.13	31.57	394.07	773.23	1056.41	1.1E+04
Short Leg	19.82	719.08	-2.3E+04	-1010.22	-721.13	-331.58	26.06	371.52	759.43	1059.00	1.2E+04
Avg. Size (Traded Stocks)	1923.6	3197.4	7.1	62.1	97.9	212.4	455.2	2390.8	5534.4	8512.0	32052.1
Turnover	0.40	0.48	0.00	0.03	0.04	0.07	0.16	0.56	1.31	1.58	2.00
Up-to-Date T-Statistic	3.97	3.70	-7.80	-1.16	-0.45	1.37	3.56	6.01	9.58	10.88	15.53
Up-to-Date Monotonicity	0.09	0.06	0.00	0.02	0.02	0.04	0.07	0.12	0.17	0.22	0.41

* 143,778 monthly factor observations from January 1926 - December 2024

II. Empirical Results

A. Cross-Sectional Factor Performance

At the cross-sectional factor level, both in-sample t-statistics and monotonicity are positively related to out-of-sample and post-publication returns. The cross-sectional regression provides further information on the predictive content of in-sample t-statistics and monotonicity for future factor returns. In the univariate specifications, both variables positively predict out-of-sample and post-publication returns. A one-standard-deviation increase in the in-sample t-statistic (3.68) is associated with 8.8 bps higher out-of-sample returns and 8.5 bps higher post-publication returns. A one-standard-deviation increase in in-sample monotonicity (0.067) is associated with 18.6 bps higher out-of-sample returns and 20.7 bps higher post-publication returns. In the joint specifications, both variables remain statistically significant, indicating that each contains incremental information about future returns. The explanatory power also increases materially: for out-of-sample returns, the R^2 rises to 0.227 from 0.037 in the t-statistic-only regression and 0.167 in the monotonicity-only regression; for post-publication returns, it rises to 0.183 from 0.024 and 0.142, respectively.

Table II

OOS / Post-Publication Returns on In-Sample T-Statistics and Monotonicity

This table reports results from regressions of out-of-sample and post-publication factor returns on in-sample t-statistics or monotonicity. T-statistic is based on the top-minus-bottom factor return spread. Monotonicity is defined as the fraction of monotonic months over the total number of months. The in-sample, out-of-sample, and post-publication periods are defined by each factor's original sample period and publication year, respectively. The sample covers January 1926 through December 2024 and is based on monthly U.S. stock return data from CRSP. Accounting data are obtained from Compustat. Standard errors are robust. t-statistics

are in parentheses. *, **, and *** denote two-tail statistical significance at the 10%, 5%, and 1% levels respectively.

VARIABLES	OOS Return			Post-Publication Return		
	REG 1	REG 2	REG 3	REG 4	REG 5	REG 6
In-Sample t-stat	2.381** (2.382)		3.045*** (3.340)	2.314** (2.104)		3.047*** (3.098)
In-Sample monotonicity		278.0*** (7.402)	298.5*** (8.521)		309.1*** (7.570)	329.6*** (8.698)
Constant	38.96*** (5.902)	21.62*** (4.659)	6.867 (0.979)	36.03*** (4.632)	15.35*** (3.275)	0.588 (0.0788)
Observations	153	153	153	153	153	153
R-squared	0.037	0.167	0.227	0.024	0.142	0.183

B. Portfolio Performance

I examine factor portfolio performance after publication. Because publication dates vary across factors, I begin the analysis in 2000, by which time 35 factors have been published. Starting the sample at this point ensures that the pool of available factors is sufficiently large, reducing the influence of extreme returns that could otherwise distort the results when the factor universe is too small.

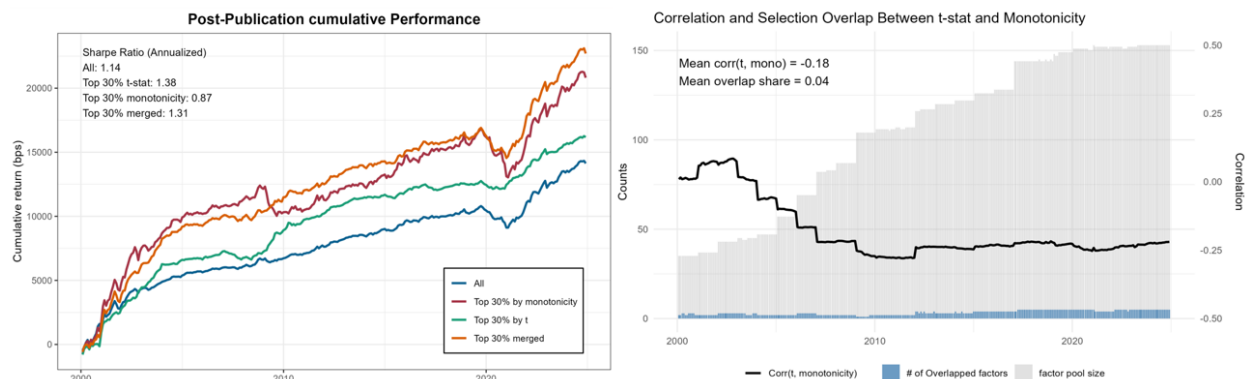
Comparison is conducted across four strategies: (1) trading all factors, (2) trading the top 30% of factors ranked by up-to-date t-statistics, (3) trading the top 30% of factors ranked by up-to-date monotonicity, and (4) trading the top 30% of factors ranked by the sum of the t-statistic rank and monotonicity rank. As it is shown in Figure III, the left panel shows that all four strategies generate positive post-publication cumulative returns. Although the monotonicity strategy has a lower Sharpe ratio than both the all-factor strategy and the t-statistic strategy (0.87, versus 1.14 and 1.38, respectively), it nevertheless delivers persistently stronger cumulative returns over time. The time-series correlation between t-statistics and monotonicity is persistently low, with a mean of -0.18, and the mean overlap share between the two selected portfolios is only 0.04. This indicates that the factors selected by the t-statistic strategy and the monotonicity strategy are largely distinct. As a result, combining the two signals allows the portfolio to exploit complementary information. Consistent with this interpretation, the merged strategy delivers the highest cumulative return by the end of the sample.

Figure III

Post-Publication Portfolio Performance across Strategies

The left panel plots cumulative returns from 2000 onward, in basis points, for four strategies: trading all published factors; trading the top 30% of factors ranked by up-to-date t-statistics; trading the top 30% ranked by up-to-date monotonicity; and trading the top 30% ranked by the sum of the t-statistic rank and monotonicity rank. Factors enter the investable universe only after publication. The right panel reports the correlation between historical t-statistics and monotonicity across published factors, together with the number of

overlapping factors selected by the two ranking rules. The t-statistic is computed from the top-minus-bottom factor return spread. Monotonicity is defined as the fraction of monotonic months out of the total number of months. The sample spans January 1926 through December 2024 and uses monthly U.S. stock return data from CRSP and accounting data from Compustat.



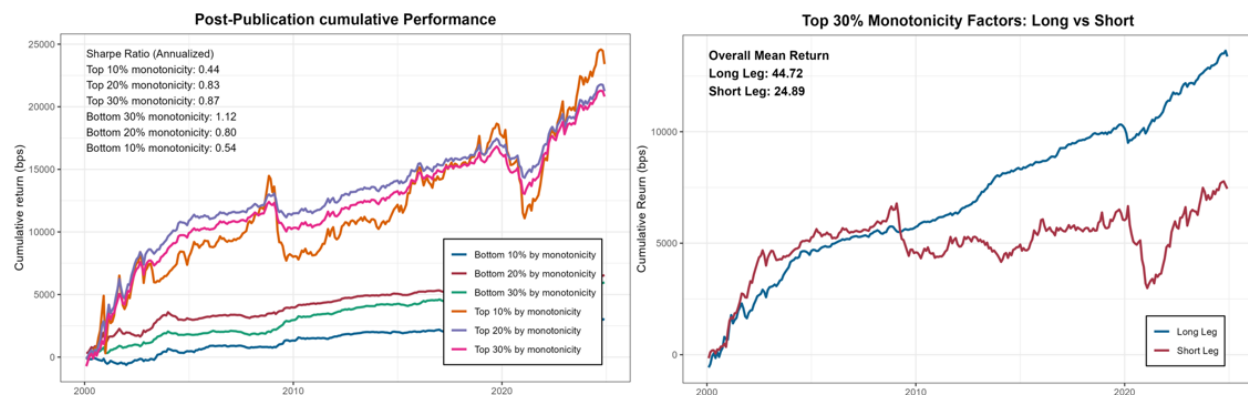
To further decompose the monotonicity strategy, the left panel of Figure IV plots cumulative returns for portfolios formed from published factors ranked by their up-to-date monotonicity. Portfolios formed from factors with higher up-to-date monotonicity deliver persistently higher cumulative returns over time. This pattern indicates that post-publication performance is concentrated among factors whose in-sample return profiles are more monotonic. At the same time, these high-monotonicity portfolios also account for most of the volatility in cumulative performance, especially for the top 10% portfolio, whose return path is more variable despite ending at the highest level. By contrast, portfolios formed from less monotonic factors earn much lower post-publication returns but exhibit relatively smooth and stable return paths. This figure indicates a tradeoff within the monotonicity sort: higher monotonicity is associated with both higher payoff potential and greater variation in realized returns, whereas lower monotonicity is associated with lower but more stable performance.

The right panel of Figure IV decomposes the top 30% monotonicity strategy into its long and short legs. The strategy's post-publication returns are persistently driven by the long leg, with the mean of 44.72 bps per month versus 24.89 bps per month from the short leg. This pattern suggests that monotonicity is effective at identifying factors, rather than relying primarily on shorting overpriced stocks. In that sense, the profitability of the monotonicity strategy appears less dependent on short-sale execution and the frictions typically associated with the short leg. The figure also shows that the short leg contributes mainly in the early part of the sample, but remains largely flat after the 2007–2009 financial crisis, whereas the long leg continues to generate substantial gains thereafter. In addition, the notable drawdown after 2020 is driven primarily by the short leg, which experiences a much sharper decline than the long leg.

Figure IV

Decomposition of Monotonicity Strategy

The left panel reports the cumulative return (bps) from 2000 onward, when trading top 10%, top 20%, top 30%, bottom 30%, bottom 20%, bottom 10% of published factors ranked by up-to-date monotonicity, respectively. The right panel reports the cumulative return (bps) of trading top 30% published factors from 2000 onward, decomposing into long leg and short leg. Monotonicity is defined as the fraction of monotonic months over the total number of months. The sample covers January 1926 through December 2024 and is based on monthly U.S. stock return data from CRSP. Accounting data are obtained from Compustat.



C. Trading Costs and Frictions

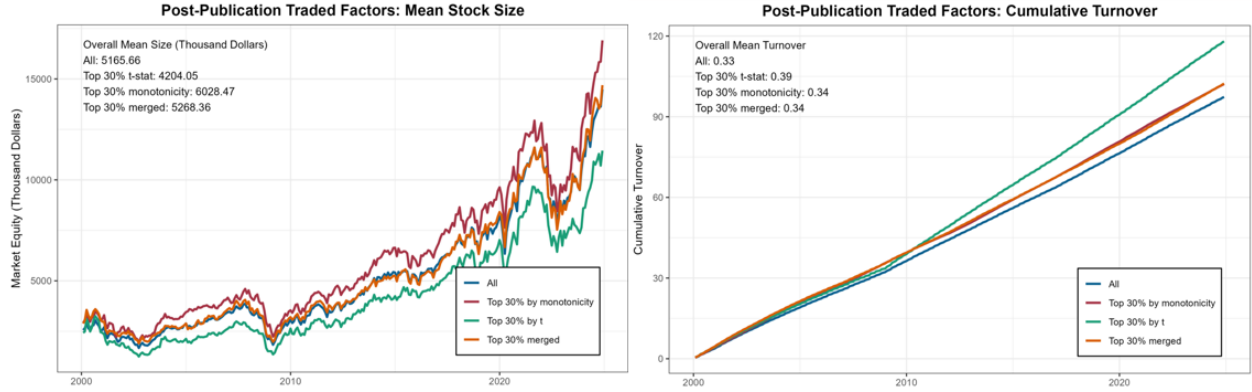
To assess whether the monotonicity strategy is likely to survive trading costs and implementation frictions, I examine the market equity and turnover of the stocks it trades. The left panel of Figure V shows that, although the average market equity of traded stocks rises over time for all strategies, the stocks selected by the monotonicity strategy are consistently larger than those in the other portfolios. This pattern suggests that the monotonicity strategy tilts toward larger, more liquid stocks with lower idiosyncratic risk, alleviating concerns that its returns are driven by hard-to-trade securities.

The right panel reports cumulative turnover. The average turnover of the monotonicity strategy is 0.34, slightly above the value for trading all factors (0.33), a difference of about 3.0%, but well below that of the t-statistic strategy (0.39), or about 12.8% lower. Thus, while the monotonicity strategy does involve somewhat more rebalancing than simply holding all factors, it turns over substantially less than the t-statistic-based strategy. Taken together, the two panels suggest that the monotonicity strategy's stronger post-publication performance is unlikely to be explained by greater exposure to trading costs. If anything, its tilt toward larger-cap stocks and its moderate turnover profile indicate that the strategy should be more implementable than strategies that rely more heavily on small, illiquid, or frequently rebalanced positions.

Figure V

Avg. Stock Market Equity and Cumulative Turnover of Traded Factors

The left panel plots, from 2000 onward, the time series of the average size of traded stocks under four strategies: trading all published factors, trading the top 30% of factors ranked by up-to-date t-statistics, trading the top 30% ranked by up-to-date monotonicity, and trading the top 30% ranked by the sum of the t-statistic rank and monotonicity rank. Factors enter the investable universe only after publication. The right panel plots the cumulative turnover of the corresponding strategies, where turnover is defined as the sum of absolute changes in stock weights for traded stocks within a factor-month. The sample covers January 1926 through December 2024 and uses monthly U.S. stock return data from CRSP and accounting data from Compustat.



III. Discussion

One may ask this question by the end of the day: how in-sample monotonicity translates into out-of-sample returns. My argument is that in-sample monotonicity improves the identification of factors that capture genuine mispricing or risk-related components, while helping to filter out spurious in-sample patterns generated by data mining.

One piece of evidence is that in-sample monotonicity is strongly associated with future monotonicity itself. Table III reports average out-of-sample and post-publication monotonicity after sorting factors by in-sample monotonicity in the cross section. The pattern is strikingly monotone. Factors in the lowest in-sample monotonicity bins exhibit almost no future monotonic structure: average out-of-sample monotonicity ranges from 0.009 to 0.032, and post-publication monotonicity from 0.011 to 0.031. By contrast, factors in the highest in-sample monotonicity bins remain strongly monotone in future periods: out-of-sample monotonicity rises to 0.153–0.244, and post-publication monotonicity to 0.149–0.242. Thus, factors in the top 5% of the in-sample monotonicity distribution are roughly twenty to twenty-seven times more monotone in future periods than factors in the bottom 5%. Because out-of-sample and post-publication monotonicity are measured in entirely different periods, this evidence suggests that monotonicity is not merely an in-sample artifact. Rather, it appears to reflect a persistent structural feature of

factor returns, consistent with the view that monotonic factors are more likely to capture genuine economic signals and therefore deliver stronger future returns.

Table III

Avg. Out-of-Sample / Post-Publication Monotonicity Sorted by In-Sample Monotonicity

This table reports the average out-of-sample and post-publication monotonicity sorted by in-sample monotonicity, at cross sectional factor level. Monotonicity is defined as the fraction of monotonic months over the total number of months. The in-sample, out-of-sample, and post-publication periods are defined by each factor's original sample period and publication year, respectively. The sample covers January 1926 through December 2024 and is based on monthly U.S. stock return data from CRSP. Accounting data are obtained from Compustat.

In-Sample Monotonicity	<5%ile	<10%ile	<20%ile	<30%ile	>70%ile	>80%ile	>90%ile	>95%ile
Out-of-Sample Monotonicity	0.009	0.021	0.024	0.032	0.153	0.162	0.208	0.244
Post-Publication Monotonicity	0.011	0.020	0.023	0.031	0.149	0.160	0.204	0.242

Implementation-wise, the top-minus-bottom strategy is most justified when returns are ordered monotonically across ranked portfolios. To illustrate this point, I compare two alternative investment schemes. In Scheme 1, the portfolio is formed by going long the top decile and short the bottom decile, corresponding to weights of

Scheme 1: $(-1, 0, 0, 0, 1)$.

In Scheme 2, the portfolio places smoother weights across the ranked portfolios, shorting two-thirds of the bottom decile and one-third of the second decile, while going long one-third of the fourth decile and two-thirds of the top decile, corresponding to weights of

Scheme 2: $(-2/3, -1/3, 0, 1/3, 2/3)$.

If out-of-sample returns are truly monotone, Scheme 1 should outperform Scheme 2, because it places all weight on the extreme portfolios where the return spread is largest.

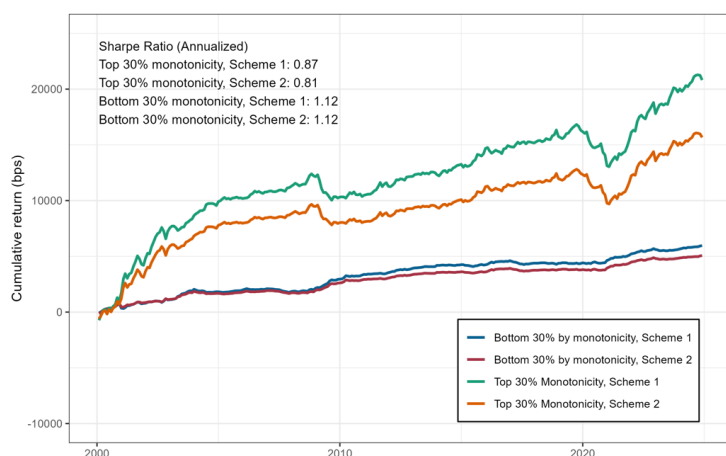
Figure VI supports this prediction. For the top 30% monotonic factors, Scheme 1 substantially outperforms Scheme 2 throughout the post-publication period, and the gap widens over time. This result indicates that for factors with strong in-sample monotonicity, the return ordering continues to hold sufficiently well out of sample that concentrating on the extreme portfolios is more effective than smoothing weights across intermediate ranks. By contrast, for the bottom 30% monotonic factors, the two schemes deliver much more similar cumulative returns and identical Sharpe ratios. This suggests that when in-sample monotonicity is weak, the monotone return structure is largely absent out of sample as well, so alternative weighting schemes across adjacent portfolios do little to improve performance. Overall, the figure provides implementation-based evidence that in-sample monotonicity is informative about the future

shape of factor returns: when monotonicity is high, the standard top-minus-bottom construction is well aligned with the underlying return structure.

Figure VI

Post-Publication Monotonicity Portfolio Performance by Schemes

This figure plots cumulative returns, in basis points, from 2000 onward for portfolios formed from the top 30% and bottom 30% of published factors ranked by up-to-date monotonicity under two alternative investment schemes. In Scheme 1, the portfolio is formed by going long the top decile and short the bottom decile, corresponding to weights of $(-1, 0, 0, 0, 1)$. In Scheme 2, the portfolio places smoother weights across the ranked portfolios, shorting two-thirds of the bottom decile and one-third of the second decile, while going long one-third of the fourth decile and two-thirds of the top decile, corresponding to weights of $(-2/3, -1/3, 0, 1/3, 2/3)$. Monotonicity is defined as the fraction of monotonic months out of the total number of months. The sample covers January 1926 through December 2024 and is based on monthly U.S. stock return data from CRSP. Accounting data are obtained from Compustat.



IV. Conclusion

While the asset pricing literature has traditionally centered on the t-statistic in identifying factors, the shape of factor return structures—specifically, monotonicity—has remained largely underexplored. This paper proposes a simple, nonparametric measure of a factor’s in-sample monotonicity: the fraction of in-sample months in which returns are non-decreasing across characteristic-sorted portfolios. I show that, in addition to the conventional t-statistic, in-sample monotonicity is a strong predictor of both out-of-sample and post-publication factor returns.

Further evidence shows that monotonicity captures a distinct dimension of factor quality from the t-statistic. Combining the two measures substantially improves the explanatory power for future returns and enhances portfolio performance. Factors with high in-sample monotonicity earn stronger subsequent returns than factors with low monotonicity, although they also

contribute more to portfolio volatility. At the same time, the profitability of monotonic factors is driven primarily by the long leg, while the short leg earns lower returns and is more vulnerable during periods of market stress. In addition, monotonicity-based strategies tend to trade stocks with larger market capitalizations and require lower turnover than strategies based on high in-sample t-statistics, suggesting that they may be more implementable in practice.

I also show that in-sample monotonicity is strongly associated with future monotonicity itself, indicating that it reflects a persistent structural feature of factor returns rather than a purely in-sample artifact. This implementation-based evidence suggests that when monotonicity is high, the standard top-minus-bottom construction is better aligned with the underlying return structure. Overall, the results suggest that monotonicity improves the identification of factors that capture genuine mispricing or risk-related components, while helping to filter out spurious in-sample patterns generated by data mining.

Finally, the baseline monotonicity measure used in this paper is intentionally simple, but it may also be overly restrictive. Because a month is classified as monotonic only when returns are strictly ordered across portfolios, the measure may understate the true extent of economically meaningful monotonicity in factor returns. One direction for future research is therefore to develop a more robust measure that captures economically meaningful monotonic structure rather than strict statistical ordering.

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